SHANZAY HUSSAIN

Aspiring Software Engineer | Full-Stack and AI-Integrated Application Developer

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Portfolio • LinkedIn • GitHub

EDUCATION

B.E. Software Engineering 2024 – 2028 (Expected)
NED University of Engineering & Technology, Karachi

FSC Pre - Engineering 2022 – 2024
Government Degree College Malir Cantt, Karachi

AREAS OF EXPERTISE

Frontend: React.js, TypeScript, JavaScript, Tailwind CSS

Backend: Node.js, Express.js, FastAPI, Django, REST APIs

Databases: MySQL, PostgreSQL, MongoDB

Programming: Python, C++, C#

AI: LLM Integration, Prompt Engineering, RAG, Embeddings, Scikit-learn

Tools & Platforms: Git, GitHub, Vercel, Railway, Linux, n8n, Docker

EXPERIENCE

Backend Intern — *ITVE Pvt Ltd* June 2026 – Aug 2026
FastAPI · MongoDB · JWT · Cloudinary

- Contributed to the backend of a FastAPI-based REST API with JWT authentication, MongoDB (via Motor) for data persistence, and Cloudinary for image upload handling.

PROJECTS

RAG Intelligence Chatbot — **Multimodal AI Knowledge Assistant**
React.js · Vite · TypeScript · FastAPI · MongoDB Atlas · ChromaDB · Firebase Auth · OpenAI (GPT-5.4-mini) · fastembed

- Built a full-stack multimodal RAG chatbot supporting document/image uploads, conversational Q&A, and persistent chat history.
- Developed a ChromaDB-based retrieval pipeline with local embeddings and similarity-threshold routing for document-grounded responses.
- Implemented image understanding and cross-conversation knowledge search, enabling retrieval across previously uploaded documents and images.

SmartBudget — **AI-Powered Personal Finance & Spending Risk Platform**
React.js · Vite · TypeScript · Express.js · Tailwind CSS · Python · FastAPI · scikit-learn · Firebase · Groq AI (LLaMA 3.3 70B) · Chart.js

- Built an end-to-end personal finance platform with budget tracking, analytics, and ML-based overspending risk prediction.
- Developed real-time spending analytics and behavioral insights based on spending trends, mood, and day of the week.
- Integrated an AI financial assistant to generate personalized insights from user spending data.

InterviewIQ — **AI-Powered Mock Interview Platform**
React.js · Vite · Express.js · Supabase · Groq AI (LLaMA 3.3 70B) · Chart.js

- Built a full-stack AI mock interview platform with automated interview interactions and performance tracking.
- Developed an analytics dashboard using Chart.js to visualize performance trends and identify skill gaps.
- Implemented speech recognition to create a more realistic interview experience.

ResumeForge AI — **Automated Resume Tailoring Platform**

React.js · TypeScript · Express.js · Firebase/Firestore · n8n · Docker

- Built a full-stack platform that automates resume tailoring based on specific job descriptions using n8n workflows.
- Designed an automation pipeline to parse resumes and job descriptions, extract requirements, and generate role-specific resumes.

CERTIFICATIONS

Google AI Professional Certificate — Coursera April 2026

Covered foundations of AI/ML, prompt engineering, and responsible AI practices.

Agents with RAG and Langchain — IBM Aug 2026 – present

Develop applications using integrated LLM, Langchain and RAG technologies.